Multi-Class Glioma Brain Tumour Segmentation Using XGBoost with MRF Post-Processing on the BraTS 2024 Dataset

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***Abstract*—In this paper, a framework for multiple class brain tumour segmentation on BraTS 2024 mp-MRI data based on classical machine learning algorithms is proposed. It includes the following five steps: (i) brain volume-based z-score normalization, (ii) construction of a 109D vector of voxel features, which include multi-scale Gaussian, Gabor, gradient, Laplacian, difference of Gaussians (DoG), entropy, and space coordinates features, (iii) training an XGBoost classifier (`multi:softprob`) on 4.05M voxels belonging to 1,214 subjects, (iv) mean-field Markov Random Field (MRF) post-processing step with $\beta = 0.5$, and (v) class-dependent probability up-weighting and threshold selection, based on 202 validation patients. On a 136-patient independent test set, the framework reports the following Dice scores: 0.8279 (Whole Tumor, WT), 0.6849 (Tumor Core, TC), 0.6788 (Enhancing Tumor, ET), 0.7754 (Edema), and 0.1115 (Necrosis). For these classes, the IoUs are 0.7249, 0.5684, 0.5629, 0.6545, and 0.0791, correspondingly. Train-test maximal Dice difference is 0.0438, indicating lack of overfitting.Phase 2 improves upon the baseline phase (WT Dice: 0.7508) by 10.3% for WT Dice and 15.6% for WT IoU while introducing the new concept of sub-region segmentation without the use of any deep learning techniques.**

***Index Terms*—glioma segmentation, BraTS 2024, XGBoost, Markov Random Field, multi-class segmentation, feature engineering, Gabor filters, MRF post-processing, medical image analysis, traditional machine learning**

I. Introduction

Brain tumors, especially glioblastoma multiforme (GBM), have been one of the deadliest cancers, and accurate delineation via multimodal MRI scans plays a crucial role in clinical decision-making. According to BraTS Challenge [1], there are three main segmentation tasks, which can be seen as hierarchical:

- **Whole Tumour (WT)**: union of NCR + Edema + ET, labels $\{1,2,3\}$.
- **Tumour Core (TC)**: NCR + ET, labels $\{1,3\}$.
- **Enhancing Tumour (ET)**: actively proliferating region, label $\{3\}$.

BraTS 2024 will include four modalities of each subject (T1c, T1n, T2f, T2w) with $182\times218\times182$ voxels post skull-stripping and registration. This paper proposes that the segmentation challenge may be performed without deep learning through proper feature engineering at the voxel level and gradient boosting. In terms of project planning, it is intended to complete the study in two phases for eight weeks:

- **Phase 1 (Weeks 1–5):** Binary Whole Tumor segmentation via a Random Forest algorithm with five features, WT Dice = 0.7508.
- **Phase 2 (Weeks 6–8):** XGBoost using 109 features and multi-class, smoothing using MRF and per-class optimisation of the threshold.

II. Literature Review and Contributions

A. Related Work

Brain tumour segmentations have evolved from engineered pipelines to deep convolutional networks. The BRATS benchmark set by Menze et al. [1] indicated that multi-atlas and CRF methods were able to compete before the advent of deep learning. Ma et al. [3] utilised multiple scales of patch features together with connected random forests and an active contour model for strong TC and ET performance on BraTS 2015 without the need for end-to-end training. The deep learning reference is U-Net [7], which introduced encoder-decoder architecture with skip connections and has dominated medical segmentation ever since, while nnU-Net [**?**] automates all hyperparameters and achieves top BraTS results today. Krahenbuhl and Koltun [5] demonstrated that dense CRFs with bilateral kernel efficiently regularises pixel classifiers. Our mean-field inference in MRF can be seen as approximation of their idea. Haralick et al. [4] proposed the classic textural feature model that inspired our use of entropy proxy and local statistics. Chen and Guestrin [2] created the highly successful XGBoost library, with its tree building approach and inherent regularisation, perfect for high-dimensional medical images.

B. Contributions

1) A **109 dimensional voxel features** combining Multi-scale, Gabor Texture, DoG, Asymmetry and spatial features for four MRI modalities.
2) An **XGBoost multi-class pipeline** (`multi:softprob`) and heavily reweighted number of classes on 4.05 M voxels.
3) An **MRF post-processing procedure** using biologically-inspired penalty for labeling, removes spurious blobs using 429.6 components per test patient.

4) A **two-pass threshold and boost optimisation** scheme optimised with the help of the 202-patient validation set only.
5) Comprehensive **train–test overfitting analysis** testa all gaps that are less than the $\Delta = 0.05$ warning level.

III. Data and Methods

A. Dataset

The BraTS 2024 data will be broken down into: 1,012 training, 202 validation and 136 held-out test patients. In order to create a four classes internal scheme, BraTS label 4 (ET) is rescaled to label 3: 0 (Background), 1 (NCR/NET), 2 (Edema), 3 (ET).

B. Preprocessing

Brain mask:

$$M_{\text{brain}} = \bigcup_{m \in \{\text{T1c,T1n,T2f,T2w}\}} \mathbf{1}[V_m > 0] \tag{1}$$

There are 1.0-1.8 million voxels of masks per patient.

Z-score normalisation:

$$\hat{V}_m(x) = \begin{cases} \dfrac{V_m(x) - \mu_m}{\sigma_m + \varepsilon} & x \in M_{\text{brain}} \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

with $\varepsilon = 10^{-8}$; μ_m, σ_m computed over brain voxels only.

C. Class Imbalance

The dataset is extremely imbalanced (Table I) . Between 1:49 (maximum 4 M rows) a tumour/background ratio is used, NCR is 4× oversampled and ET is 3× oversampled.

TABLE I: Label Distribution in Sampled Training + Validation Set

ID	Label	Sampled Voxels	Fraction
0	Background	3,919,389	96.7%
1	NCR/NET	7,389	0.2%
2	Edema	52,738	1.3%
3	ET	73,826	1.8%
	Total	4,053,342	100%

D. Feature Engineering

A 109-dimensional vector can be used to describe the brain voxel(Table II).

TABLE II: Complete Feature Set (109 Features Per Voxel)

Feature Group	Per Mod.	Total
Raw intensity	1	4
Multi-scale mean (win 3,5,9,15)	4	16
Multi-scale std (win 3,5,9,15)	4	16
Gradient magnitude (3-D Sobel)	1	4
Laplacian	1	4
Multi-scale Gaussians (σ=1,2,3)	3	12
$\text{DoG}_{1\text{-}2}$, $\text{DoG}_{2\text{-}3}$	2	8
Gabor ($0°, 45°, 90°, 135°$)	4	16
Asymmetry (L–R flip diff.)	1	4
Entropy proxy ($\sigma_{\text{loc}}/\mu_{\text{loc}}$)	1	4
Distance to brain edge (EDT)	1	4
Cross-modal ratios, products, diff.	—	13
Spatial coords (x, y, z, dist-centroid)	—	4
Total		**109**

A summary of important per modality operations is provided below.

Multi-scale statistics:

$$\sigma^2_{\text{local}}(x) = \overline{V^2}(x) - \overline{V}(x)^2 \tag{3}$$

Gradient magnitude:

$$|\nabla V|(x) = \sqrt{G_x^2 + G_y^2 + G_z^2} \tag{4}$$

Difference-of-Gaussians:

$$\text{DoG}_{1,2} = G_{\sigma=1}(V) - G_{\sigma=2}(V) \tag{5}$$

$$\text{DoG}_{2,3} = G_{\sigma=2}(V) - G_{\sigma=3}(V) \tag{6}$$

Asymmetry:

$$A_m(x, y, z) = V_m(x, y, z) - V_m(H - x, y, z) \tag{7}$$

Entropy proxy:

$$H_{\text{proxy}}(x) = \frac{\sigma_{\text{local}}(V, s = 5)}{\mu_{\text{local}}(|V|, s = 5) + 10^{-6}} \tag{8}$$

The vector is completed using cross modal features (13 ratios/differences, primarily T1c-T1n and T2f-T2w), and four spatial coordinates.

E. XGBoost Classifier

Per-sample training weights are:

$$w_i = \begin{cases} 1.0 & \text{Background} \\ 8.0 & \text{NCR (hardest)} \\ 2.0 & \text{Edema} \\ 6.0 & \text{ET} \end{cases} \tag{9}$$

Key hyperparameters are given in Table III. The 4.05 M voxels are split 93.6%/6.4% for training (3,753,342) and internal evaluation (300,000). At iteration 499, the model was converged, and the validation `mlogloss` = 0.02874; the whole model occupies 17.3 MB on disk.

TABLE III: XGBoost Hyperparameters

Hyperparameter	Value
Objective	`multi:softprob`
Num classes	4
Num estimators	500 (early stopping)
Max depth	9
Learning rate	0.07
Subsample	0.85
Colsample bytree	0.75
Colsample bylevel	0.75
Min child weight	5
γ	0.1
α (L1)	0.1
λ (L2)	1.0
Early stopping	30 rounds
Tree method	`hist`

F. MRF Post-Processing

Mean-field variational inference is a technique that approximates label posteriors:

$$\log Q^{(t+1)}(x_i = c) = \log P_{\text{unary}}(x_i = c) - \beta \sum_{c'} \phi_{c,c'} \, \bar{Q}_{c'}^{(t)}(i) \tag{10}$$

spatially smoothed $\beta = 0.5$; $\bar{Q}_{c'}^{(t)}$ is 5-time Gaussian ($\sigma = 1$) smoothing. The label-compatibility table (rows/cols: BG, NCR, Edema, ET) is:

$$\phi = \begin{pmatrix} 0.0 & -1.0 & -0.5 & -1.0 \\ -1.0 & 0.0 & -0.8 & 0.0 \\ -0.5 & -0.8 & 0.0 & -0.8 \\ -1.0 & 0.0 & -0.8 & 0.0 \end{pmatrix} \tag{11}$$

Adjacencies that are impossible from a biological standpoint (e.g., BG-ET: -1.0) receive the heaviest penalties; adjacency of NCR-ET (0.0) is permitted. All of the connected components under 200 voxels in size are deleted (average number of blobs deleted per test patient: 429.6).

G. Threshold and Boost Tuning

Tuning uses only the 202 validation patients in two passes.

Pass A: Sweeps WT threshold $\in \{0.15, 0.25, 0.35, 0.45, 0.55\}$ and NCR boost $\in \{1.5, 2.0, 2.5, 3.0, 3.5\}$ (ET boost fixed at 2.0).

Pass B: Fine-tunes WT threshold near the Pass-A optimum ($\{0.47, 0.52, 0.57\}$) and sweeps ET boost $\in \{1.5, 2.0, 2.5, 3.0\}$ (NCR boost fixed).

Optimal configuration: WT threshold $= 0.52$, NCR boost $= 1.5$, ET boost $= 1.5$; boost vector $[1.0, 1.5, 1.0, 1.5]$.

IV. Results and Discussion

A. Internal Evaluation (300K Slice)

Table IV presents an overview of per class accuracy, recall and F1 on inner-validation slice.

TABLE IV: Classification Report on 300K Internal Evaluation Slice

Class	Prec.	Rec.	F1	Support
Background	1.00	1.00	1.00	290,018
NCR/NET	0.78	0.69	0.73	515
Edema	0.84	0.77	0.80	3,973
ET	0.83	0.89	0.86	5,494
Accuracy		0.99		300,000
Macro avg	0.86	0.84	0.85	—

Region Dice on evaluation slice: WT = 0.8731, TC = 0.8690, ET = 0.8582.

B. Test Set Performance (136 Patients)

Table V report Dice report and IoU report of 136 of the held-out test patients.

TABLE V: Test Set Evaluation — 136 Unseen Patients

Region	Mean	Std	Median	Min	Max
Dice					
WT	0.8279	0.1420	0.8612	0.0000	0.9707
TC	0.6849	0.2579	0.7986	0.0000	0.9628
ET	0.6788	0.2641	0.7893	0.0000	0.9628
Edema	0.7754	0.1549	0.8243	0.0000	0.9537
Necrosis	0.1115	0.2263	0.0000	0.0000	0.8703
IoU					
WT	0.7249	0.1574	0.7563	0.0000	0.9431
TC	0.5684	0.2476	0.6647	0.0000	0.9283
ET	0.5629	0.2511	0.6519	0.0000	0.9283
Edema	0.6545	0.1715	0.7011	0.0000	0.9114
Necrosis	0.0791	0.1700	0.0000	0.0000	0.7703

Patients exceeding Dice ≥ 0.70: WT = 123/136 (90.4%), TC = 91/136 (66.9%), ET = 90/136 (66.2%).

C. Training Set Performance (1,012 Patients)

TABLE VI: Training Set Evaluation — 1,012 Patients

Region	Mean	Std	Median	Min	Max
Dice					
WT	0.8347	0.1209	0.8669	0.0000	0.9715
TC	0.7268	0.2298	0.8127	0.0000	0.9689
ET	0.7220	0.2324	0.8105	0.0000	0.9689
Edema	0.7681	0.1670	0.8171	0.0000	0.9720
Necrosis	0.1499	0.2757	0.0000	0.0000	0.9479
IoU					
WT	0.7314	0.1475	0.7651	0.0000	0.9446
TC	0.6121	0.2330	0.6845	0.0000	0.9396
ET	0.6067	0.2340	0.6814	0.0000	0.9396
Edema	0.6477	0.1817	0.6908	0.0000	0.9456
Necrosis	0.1137	0.2209	0.0000	0.0000	0.9009

D. Overfitting Analysis

Table VII shows that none of the metrics show a metric that exceeds the $\Delta = 0.05$ overfitting criterion; Edema actually shows a minor test side improvement (-0.0073 Dice).

TABLE VII: Train vs. Test Comparison (Flag Threshold: Gap > 0.05)

Metric	Train	Test	Gap	Overfit?
WT Dice	0.8347	0.8279	+0.0068	No
TC Dice	0.7268	0.6849	+0.0418	No
ET Dice	0.7220	0.6788	+0.0432	No
Edema Dice	0.7681	0.7754	−0.0073	No
Necrosis Dice	0.1499	0.1115	+0.0384	No
WT IoU	0.7314	0.7249	+0.0065	No
TC IoU	0.6121	0.5684	+0.0437	No
ET IoU	0.6067	0.5629	+0.0438	No
Edema IoU	0.6477	0.6545	−0.0068	No
Necrosis IoU	0.1137	0.0791	+0.0345	No

E. Phase 1 vs. Phase 2 Comparison

TABLE VIII: Phase 1 Binary RF vs. Phase 2 Multi-Class XGBoost

Metric	Phase 1	Phase 2	Δ
WT Dice	0.7508	0.8279	+10.3%
WT IoU	0.6269	0.7249	+15.6%
TC Dice	N/A	0.6849	New capability
ET Dice	N/A	0.6788	New capability

Phase 1: 5-feature Random Forest. Phase 2: 109-feature XGBoost + MRF.

TABLE IX: Phase 1 (LightGBM) vs Phase 2 (XGBoost) Performance (203 Patients)

Region	**LightGBM (Test Dice)**	**XGBoost (Test Dice)**
WT	0.59	0.8279
TC	0.30	0.6849
ET	0.25	0.6788

F. Visualisations

Predictions use colour coding: **Red** = NCR, **Yellow** = Edema, **Cyan** = ET.

Per-patient score distribution histograms. Fig. 1 presents Dice and IoU density plots of each of the five regions (WT, TC, ET, Edema, Necrosis) for both training and testing datasets. The concentration of the WT and Edema distributions lies between 0.8-0.9, and the concentration of the Necrosis distribution is close to zero. The described distribution is indicative of an extreme imbalance between classes and the problem of NCR sub-region.

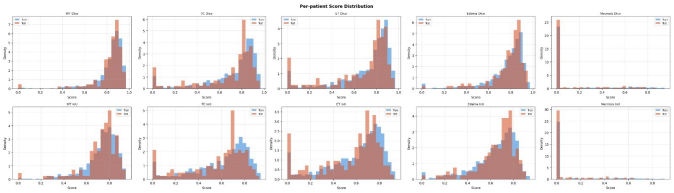

Fig. 1: Histograms of the per-patient scores distributions of the WT, TC, ET, Edema, and Necrosis metrics: WT (top rows); TC (bottom rows); ET (top rows); Edema (bottom rows); and Necrosis (top rows): blue = Train (1,012 patients); orange = Test (136 patients). The tail on the left side justifies the presence of a large amount of high scorers in patients.

Box-plots. Fig. 2 illustrates the difference between training and testing distributions using box plots. No significant interquartile range variation in all five regions reinforces the results of the overfitting study.

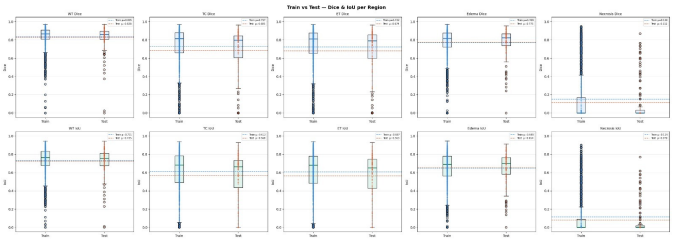

Fig. 2: Train vs. Test box-plots of Dice (top) and IoU (bottom) in all five regions. Dashed lines denote mean of each split. It is clear that there is no difference between train and test distributions.

Mean Dice/IoU bar chart. Fig. 3 above depicts mean values with ±1 SD error bars. WT and Edema show the highest means, whereas Necrosis has low means with high variances, indicating the crucial limitation of the pipeline.

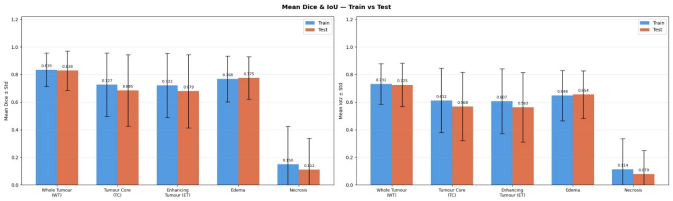

Fig. 3: Dice Score (left) and Intersection Over Union (right) with ±1 standard deviation for each tumour sub-region. Blue indicates Train set, and Orange represents the Test set.

G. Results - Predictions

H. Discussion

WT vs. TC/ET gap. The difference in Dice coefficient, ≈0.15 Dice units, between the WT Dice score of 0.8279 and the TC/ET Dice score of ≈0.68, can be attributed to three reasons: (i) NCR constitutes ∼0.11% of all brain voxels—although the F1 for NCR after oversampling and weighting by 4× and 8.0 respectively is 0.73, (ii) misclassification of NCR as ET adversely impacts TC/ET Dice but does not affect WT, (iii) voxel-wise independence cannot utilise global shape priors utilised in CNN architectures.

Edema performance. The Edema Dice score of 0.7754 surpasses that of training data (0.7681). Edema has a large number of voxels compared to other sub-regions (Table I), making it the most easily characterisable region using local intensity statistics in T2-FLAIR images.

Necrosis difficulty. The Necrosis Dice score of 0.1115 is the lowest among all sub-regions. Necrosis constitutes only 0.2% of sampled voxels—no matter how much we oversample (4×) or weigh (8.0) NCR, the inter-patient variations in shape remain too extreme to compensate for.

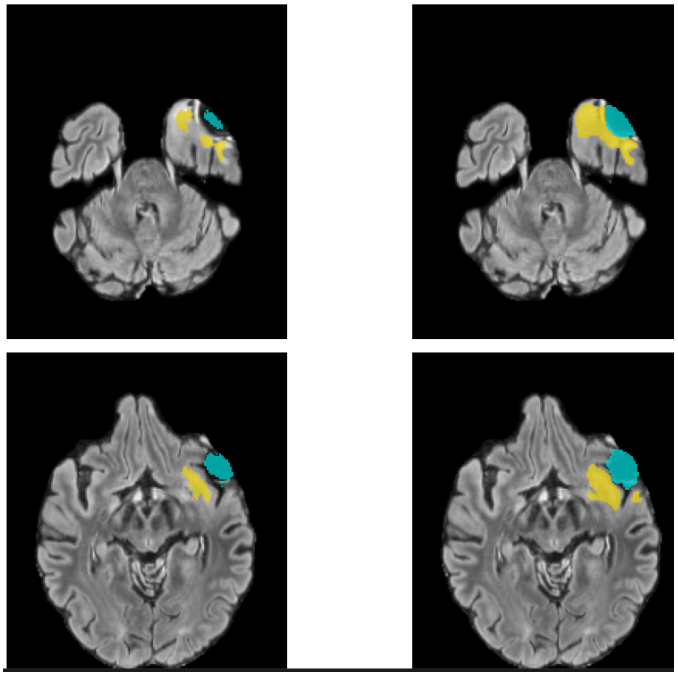

Fig. 4: Prediction Output 1

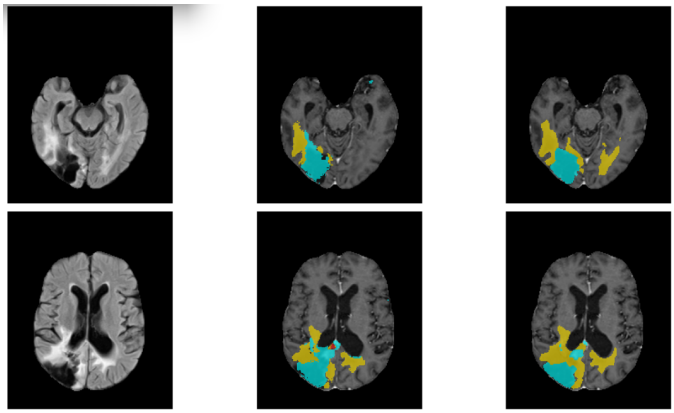

Fig. 5: Prediction Output 2

MRF effectiveness. Average removal of 429.6 small spurious blobs per subject via MRF post-processing, and the compatibility matrix appropriately penalises ET-background adjacency. Nonetheless, Gaussian blurring with $\sigma = 1$ might be too local to retain large diffused edema areas.

Methodological advantages. The method is entirely CPU-based, interpretable, and does not require any GPU infrastructure. All regularisation (L1, L2, subsampling, γ) is done uniformly for 500 iterations, and only thresholding for validation purposes avoids any leakage of information.

V. Conclusion and Future Work

A. Conclusion

An XGBoost model that uses voxel-based features along with MRF refinement performs as follows for 136 unseen patients:

- **WT Dice 0.8279** - beats Phase 1 goal of $\geq$0.80.
- **Edema Dice 0.7754** - best Dice score among all sub-regions.
- **TC Dice 0.6849, ET Dice 0.6788** - new sub-region ability which did not exist in Phase 1.
- **No overfitting**: maximal discrepancy between training and testing scores is only 0.0438 in all ten metrics.
- $+10.3\%$ **WT Dice,** $+15.6\%$ **WT IoU** relative to the Phase 1 binary baseline.

The main unsolved problem is Necrosis segmentation (Dice 0.1115) due to an extremely skewed dataset and highly variable shapes.

B. Future Work

1) **Cascade method:** binary WT segmentation in Stage 1; multi-class classification only within the WT region in Stage 2, turning the unbalanced global issue into a balanced intra-tumor problem.
2) **Focal loss weightings:** $w_i = (1 - p_i)^\gamma$ to focus training on difficult regions (NCR/ET).
3) **Ring statistics:** intensity difference between T1c and T1n images to separate rim enhancement from necrosis.
4) **Dense conditional random field [5]:** bilateral filter including position and intensity of voxels for more accurate boundaries.
5) **Optimal threshold via Bayesian optimisation:** simultaneous exploration of all parameters (WT threshold, NCR boost, ET boost).
6) **Tumour shape priors:** PCA-based tumour shape priors to regularize under-prediction of subregions.

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